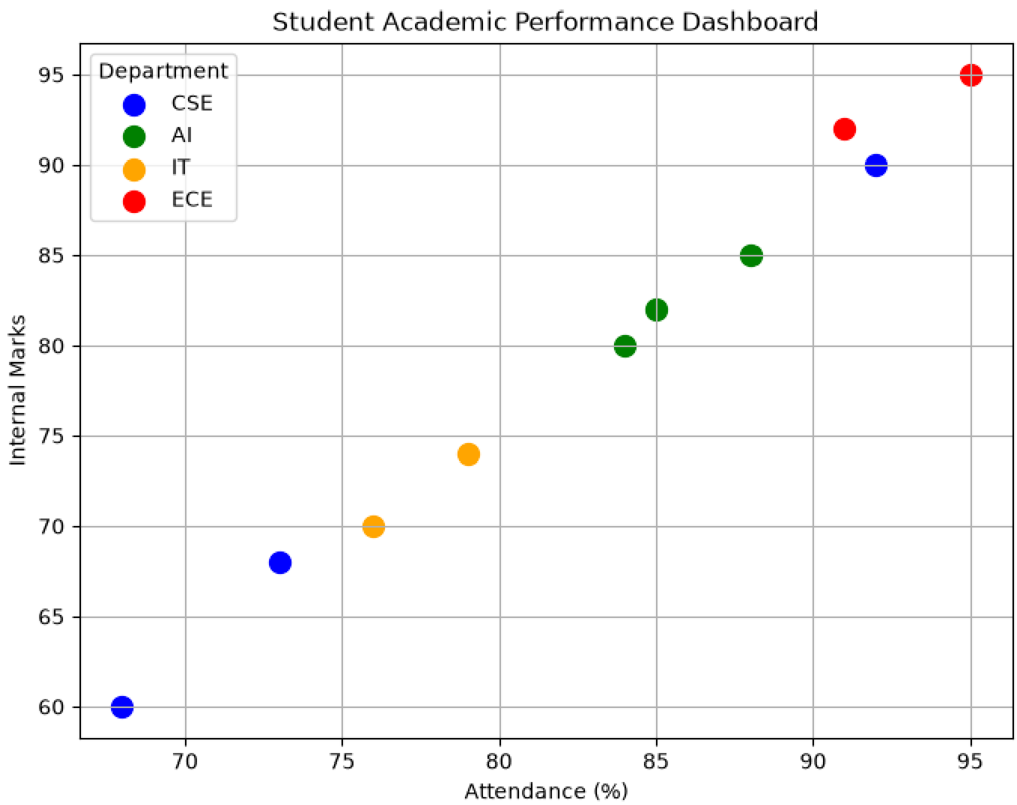
Student Academic Performance Dashboard

Program:

```
import pandas as pd
import matplotlib.pyplot as plt

# -----------------------------
# Sample Dataset
# -----------------------------
data = {
    "Student ID": [101, 102, 103, 104, 105, 106, 107, 108, 109, 110],
    "Attendance": [92, 85, 76, 95, 68, 88, 79, 91, 73, 84],
    "Internal Marks": [90, 82, 70, 95, 60, 85, 74, 92, 68, 80],
    "Department": ["CSE", "AI", "IT", "ECE", "CSE", "AI", "IT", "ECE", "CSE", "AI"]
}
df = pd.DataFrame(data)

# -----------------------------
# Define colors for departments
# -----------------------------
colors = {
    "CSE": "blue",
    "AI": "green",
    "IT": "orange",
    "ECE": "red"
}

# -----------------------------
# Scatter Plot
# -----------------------------
plt.figure(figsize=(8, 6))

# Plot each department separately
for dept in df["Department"].unique():
    dept_data = df[df["Department"] == dept]
    plt.scatter(
        dept_data["Attendance"],
        dept_data["Internal Marks"],
        color=colors[dept],
        label=dept,
        s=100
    )

# -----------------------------
# Chart Customization
# -----------------------------
plt.title("Student Academic Performance Dashboard")
plt.xlabel("Attendance (%)")
plt.ylabel("Internal Marks")
```

```
plt.legend(title="Department")
plt.grid(True)

# Display the plot
plt.show()
```

Output:



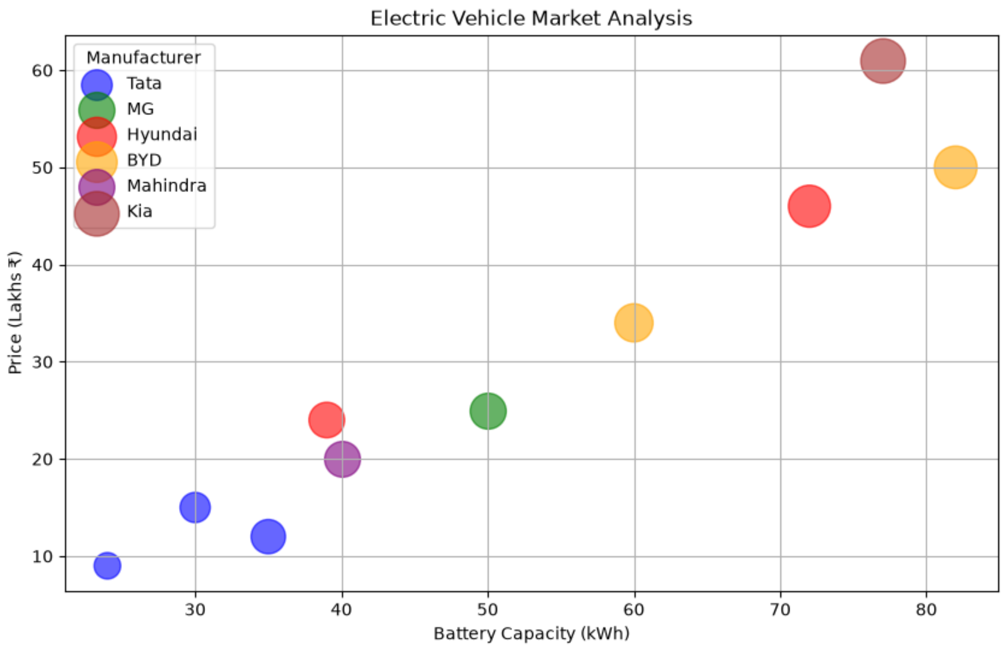
Electric Vehicle Market Analysis

Program:

```
import pandas as pd
import matplotlib.pyplot as plt

# ----------------------------
# Sample Dataset
# ----------------------------
data = {
    "Vehicle Model": [
        "Nexon EV", "MG ZS EV", "Tiago EV", "Kona Electric",
        "BYD Atto 3", "Mahindra XUV400", "Tata Punch EV",
        "Hyundai Ioniq 5", "BYD Seal", "Kia EV6"
    ],
    "Battery Capacity": [30, 50, 24, 39, 60, 40, 35, 72, 82, 77],
    "Price": [15, 25, 9, 24, 34, 20, 12, 46, 50, 61],   # Price in Lakhs
    "Driving Range": [325, 461, 250, 452, 521, 456, 421, 631, 650, 708],
```

```
    "Manufacturer": [
        "Tata", "MG", "Tata", "Hyundai",
        "BYD", "Mahindra", "Tata",
        "Hyundai", "BYD", "Kia"
    ]
}

df = pd.DataFrame(data)

# ----------------------------
# Assign Colors to Manufacturers
# ----------------------------
colors = {
    "Tata": "blue",
    "MG": "green",
    "Hyundai": "red",
    "BYD": "orange",
    "Mahindra": "purple",
    "Kia": "brown"
}

# ----------------------------
# Bubble Chart
# ----------------------------
plt.figure(figsize=(10, 6))

for company in df["Manufacturer"].unique():
    company_data = df[df["Manufacturer"] == company]

    plt.scatter(
        company_data["Battery Capacity"],
        company_data["Price"],
        s=company_data["Driving Range"],   # Bubble size
        color=colors[company],
        alpha=0.6,
        label=company
    )

# ----------------------------
# Chart Customization
# ----------------------------
plt.title("Electric Vehicle Market Analysis")
plt.xlabel("Battery Capacity (kWh)")
plt.ylabel("Price (Lakhs ₹)")
plt.legend(title="Manufacturer")
plt.grid(True)

plt.show()
```

Output:



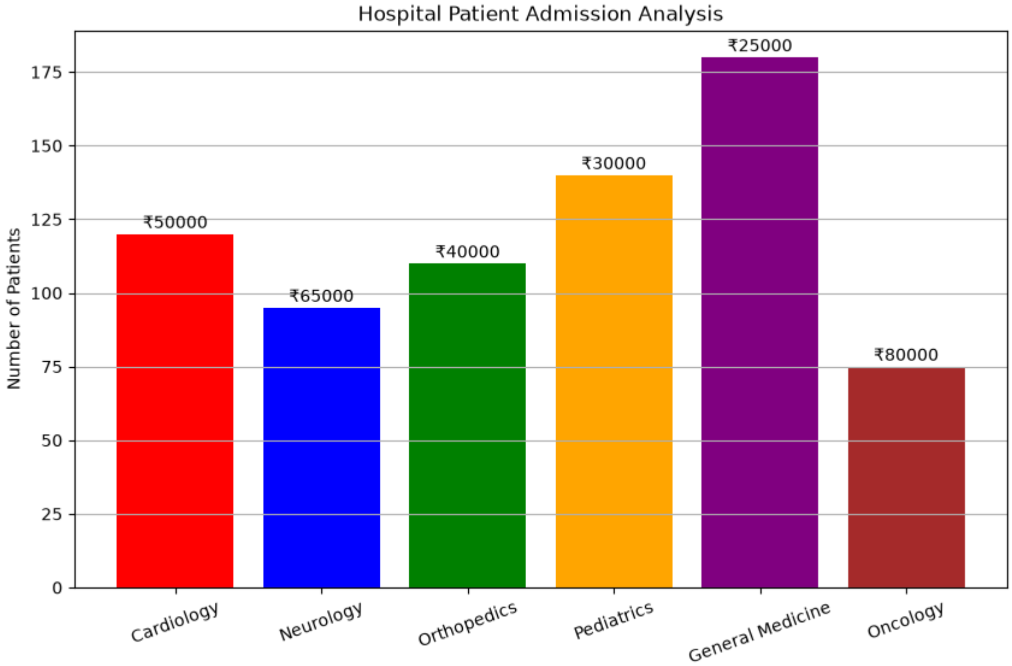
Hospital Patient Admission Analysis

Program:

```
import pandas as pd
import matplotlib.pyplot as plt

# ----------------------------
# Sample Dataset
# ----------------------------
data = {
    "Department": [
        "Cardiology",
        "Neurology",
        "Orthopedics",
        "Pediatrics",
        "General Medicine",
        "Oncology"
    ],
    "Number of Patients": [120, 95, 110, 140, 180, 75],
    "Average Treatment Cost": [50000, 65000, 40000, 30000, 25000, 80000]
}

df = pd.DataFrame(data)

# ----------------------------
# Colors for Departments
```

```
# -----------------------------
colors = [
    "red",
    "blue",
    "green",
    "orange",
    "purple",
    "brown"
]

# -----------------------------
# Bar Chart
# -----------------------------
plt.figure(figsize=(10, 6))

bars = plt.bar(
    df["Department"],
    df["Number of Patients"],
    color=colors
)

# -----------------------------
# Add Treatment Cost Labels
# -----------------------------
for bar, cost in zip(bars, df["Average Treatment Cost"]):
    plt.text(
        bar.get_x() + bar.get_width()/2,
        bar.get_height() + 2,
        f"₹{cost}",
        ha="center",
        fontsize=10
    )

# -----------------------------
# Chart Customization
# -----------------------------
plt.title("Hospital Patient Admission Analysis")
plt.xlabel("Department")
plt.ylabel("Number of Patients")
plt.xticks(rotation=20)
plt.grid(axis="y")

plt.show()
```

Output:



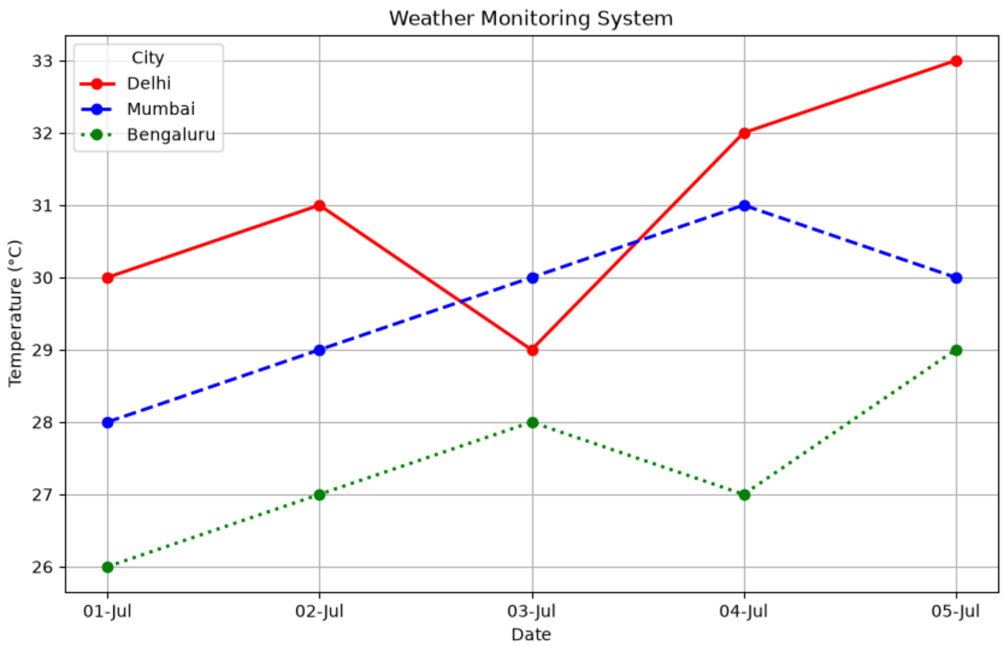
Weather Monitoring System

Program:

```
import pandas as pd
import matplotlib.pyplot as plt

# ----------------------------
# Sample Dataset
# ----------------------------
data = {
    "Date": [
        "01-Jul", "02-Jul", "03-Jul", "04-Jul", "05-Jul",
        "01-Jul", "02-Jul", "03-Jul", "04-Jul", "05-Jul",
        "01-Jul", "02-Jul", "03-Jul", "04-Jul", "05-Jul"
    ],
    "Temperature": [
        30, 31, 29, 32, 33,     # Delhi
        28, 29, 30, 31, 30,     # Mumbai
        26, 27, 28, 27, 29      # Bengaluru
    ],
    "City": [
        "Delhi", "Delhi", "Delhi", "Delhi", "Delhi",
        "Mumbai", "Mumbai", "Mumbai", "Mumbai", "Mumbai",
        "Bengaluru", "Bengaluru", "Bengaluru", "Bengaluru", "Bengaluru"
    ]
}

df = pd.DataFrame(data)

# ----------------------------
# Colors and Line Styles
```

```
# ----------------------------
colors = {
    "Delhi": "red",
    "Mumbai": "blue",
    "Bengaluru": "green"
}

line_styles = {
    "Delhi": "-",
    "Mumbai": "--",
    "Bengaluru": ":"
}

# ----------------------------
# Line Chart
# ----------------------------
plt.figure(figsize=(10, 6))

for city in df["City"].unique():
    city_data = df[df["City"] == city]

    plt.plot(
        city_data["Date"],
        city_data["Temperature"],
        color=colors[city],
        linestyle=line_styles[city],
        linewidth=2,
        marker="o",
        label=city
    )

# ----------------------------
# Chart Customization
# ----------------------------
plt.title("Weather Monitoring System")
plt.xlabel("Date")
plt.ylabel("Temperature (°C)")
plt.legend(title="City")
plt.grid(True)

plt.show()
```

Output:



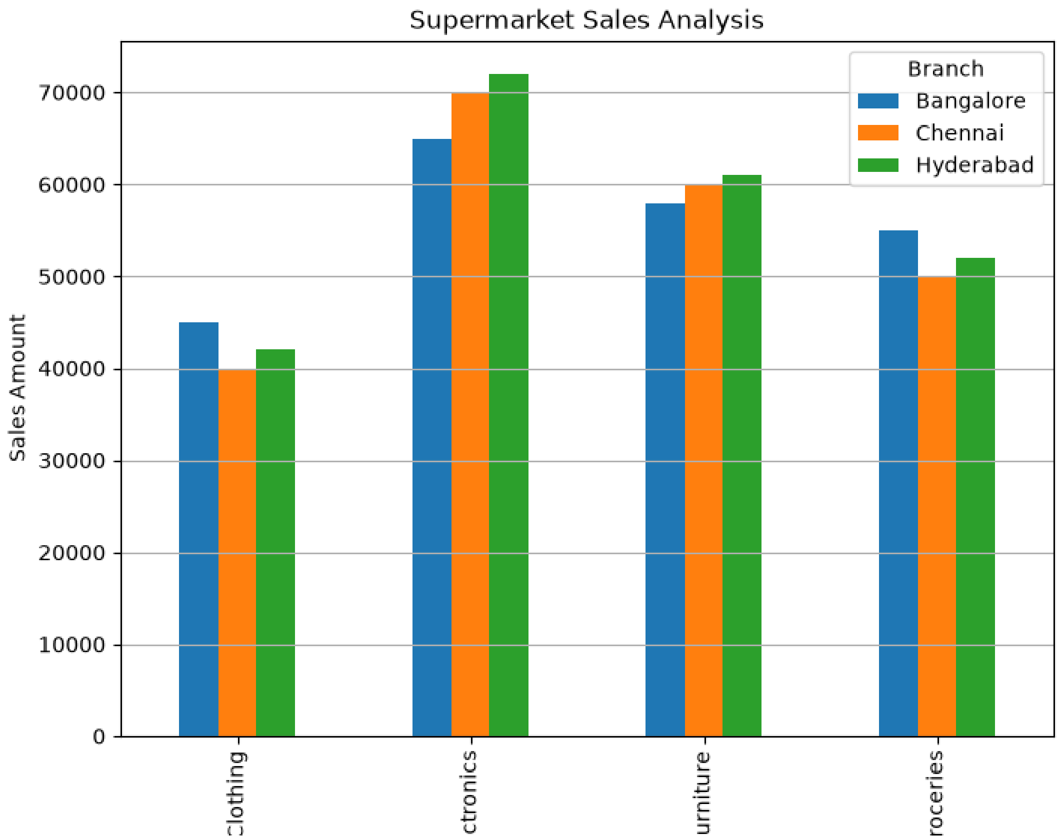
Supermarket Sales Analysis

Program:

```
import pandas as pd
import matplotlib.pyplot as plt
import numpy as np

# ----------------------------
# Sample Dataset
# ----------------------------
data = {
    "Month": ["January"] * 12,
    "Product Category": [
        "Groceries", "Electronics", "Clothing", "Furniture",
        "Groceries", "Electronics", "Clothing", "Furniture",
        "Groceries", "Electronics", "Clothing", "Furniture"
    ],
    "Sales Amount": [
        50000, 70000, 40000, 60000,
        55000, 65000, 45000, 58000,
        52000, 72000, 42000, 61000
    ],
    "Branch": [
        "Chennai", "Chennai", "Chennai", "Chennai",
        "Bangalore", "Bangalore", "Bangalore", "Bangalore",
        "Hyderabad", "Hyderabad", "Hyderabad", "Hyderabad"
```

```
    ]
}

df = pd.DataFrame(data)

# Pivot table
pivot = df.pivot(index="Product Category",
                 columns="Branch",
                 values="Sales Amount")

# Plot grouped bar chart
pivot.plot(kind="bar", figsize=(8,6))

plt.title("Supermarket Sales Analysis")
plt.xlabel("Product Category")
plt.ylabel("Sales Amount")
plt.legend(title="Branch")
plt.grid(axis='y')

plt.show()
```

Output:



Employee Salary Distribution

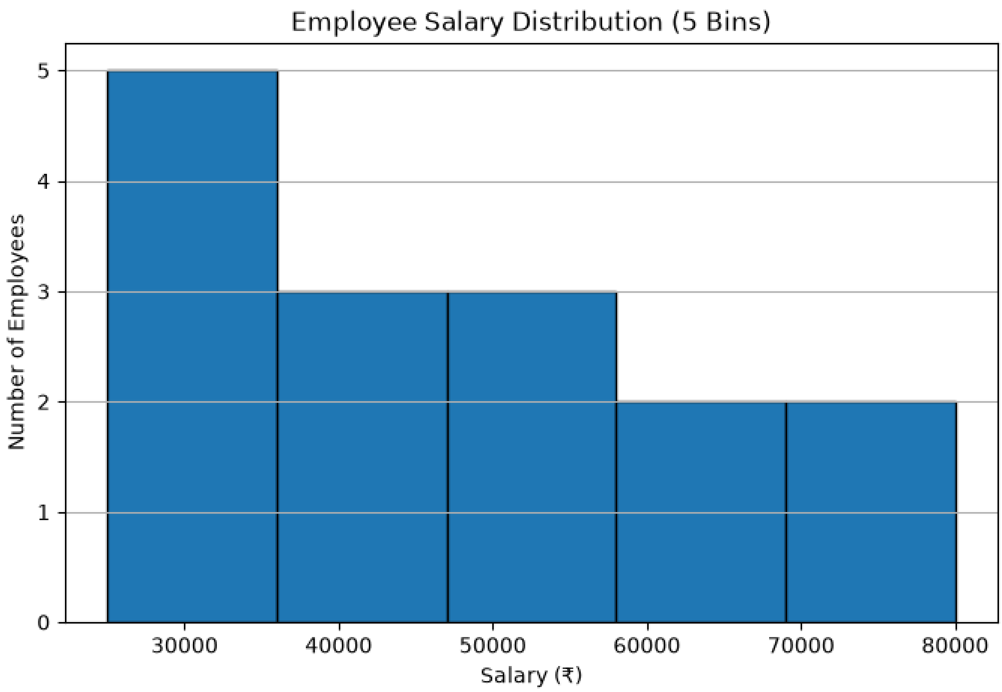
Program:

```
import pandas as pd
import matplotlib.pyplot as plt

# -----------------------------
# Sample Dataset
# -----------------------------
data = {
    "Employee ID": [101,102,103,104,105,106,107,108,109,110,
                    111,112,113,114,115],
    "Salary": [25000,28000,30000,32000,35000,
               38000,42000,45000,48000,50000,
               55000,60000,65000,70000,80000],
    "Experience": [1,2,2,3,4,5,6,7,8,9,10,11,12,13,15],
    "Department": [
        "HR","IT","Finance","IT","Sales",
        "HR","Finance","IT","Sales","HR",
        "Finance","IT","Sales","HR","Finance"
    ]
}

df = pd.DataFrame(data)

# -----------------------------
# Histogram with 5 bins
# -----------------------------
plt.figure(figsize=(8,5))

plt.hist(df["Salary"],
         bins=5,
         edgecolor="black")

plt.title("Employee Salary Distribution (5 Bins)")
plt.xlabel("Salary (₹)")
plt.ylabel("Number of Employees")

plt.grid(axis="y")

plt.show()

# -----------------------------
# Histogram with 10 bins
# -----------------------------
plt.figure(figsize=(8,5))

plt.hist(df["Salary"],
         bins=10,
         edgecolor="black")

plt.title("Employee Salary Distribution (10 Bins)")
plt.xlabel("Salary (₹)")
plt.ylabel("Number of Employees")
```

plt.grid(axis="y")

plt.show()

Output:



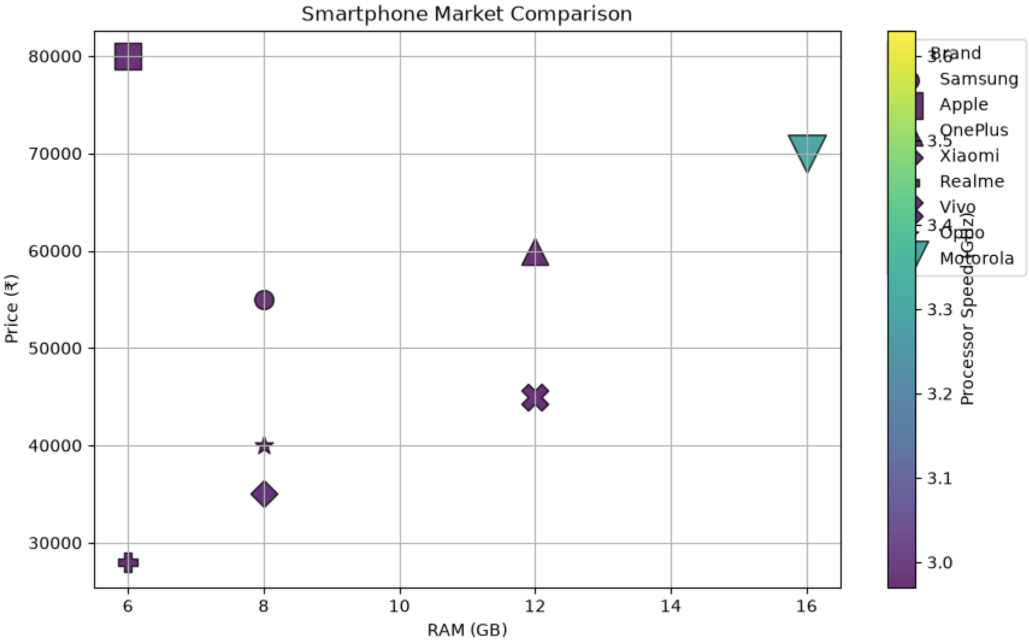
Smartphone Market Comparison

Program:

```
import pandas as pd
import matplotlib.pyplot as plt

# -----------------------------
# Sample Dataset
# -----------------------------
data = {
    "Brand": [
        "Samsung", "Apple", "OnePlus", "Xiaomi",
        "Realme", "Vivo", "Oppo", "Motorola"
    ],
    "RAM": [8, 6, 12, 8, 6, 12, 8, 16],                  # GB
    "Processor Speed": [2.8, 3.2, 3.1, 2.9, 2.7, 3.0, 2.8, 3.3],  # GHz
    "Price": [55000, 80000, 60000, 35000, 28000, 45000, 40000, 70000],  # ₹
    "Storage": [128, 256, 256, 128, 128, 256, 128, 512]  # GB
}

df = pd.DataFrame(data)

# -----------------------------
# Shapes for each Brand
# -----------------------------
```

```
markers = {
    "Samsung": "o",
    "Apple": "s",
    "OnePlus": "^",
    "Xiaomi": "D",
    "Realme": "P",
    "Vivo": "X",
    "Oppo": "*",
    "Motorola": "v"
}

# -----------------------------
# Scatter Plot
# -----------------------------
plt.figure(figsize=(10, 6))

for brand in df["Brand"].unique():
    brand_data = df[df["Brand"] == brand]

    plt.scatter(
        brand_data["RAM"],
        brand_data["Price"],
        s=brand_data["Storage"],               # Point Size
        c=brand_data["Processor Speed"],       # Color
        cmap="viridis",
        marker=markers[brand],
        edgecolors="black",
        alpha=0.8,
        label=brand
    )

# -----------------------------
# Color Bar
# -----------------------------
cbar = plt.colorbar()
cbar.set_label("Processor Speed (GHz)")

# -----------------------------
# Chart Customization
# -----------------------------
plt.title("Smartphone Market Comparison")
plt.xlabel("RAM (GB)")
plt.ylabel("Price (₹)")
plt.legend(title="Brand", bbox_to_anchor=(1.05, 1), loc="upper left")
plt.grid(True)

plt.show()
```

Output:



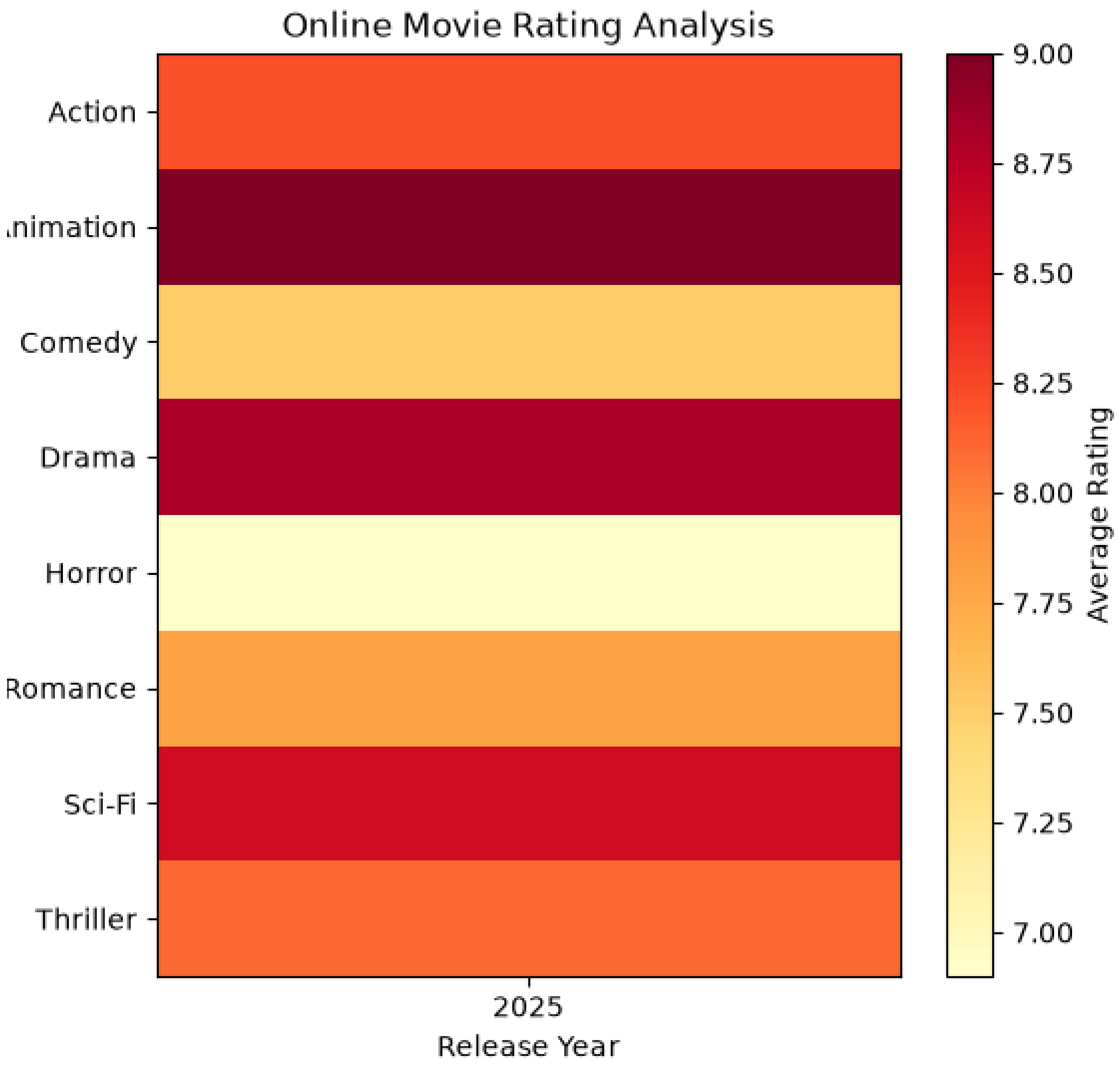
Online Movie Rating Analysis

Program:

```
import pandas as pd
import matplotlib.pyplot as plt

# ----------------------------
# Sample Dataset
# ----------------------------
data = {
    "Genre": [
        "Action", "Comedy", "Drama", "Horror",
        "Sci-Fi", "Romance", "Thriller", "Animation"
    ],
    "Average Rating": [8.2, 7.5, 8.8, 6.9, 8.6, 7.8, 8.1, 9.0],
    "Number of Reviews": [1200, 900, 1500, 700, 1300, 850, 1100, 1400],
    "Release Year": [2025, 2025, 2025, 2025, 2025, 2025, 2025, 2025]
}

df = pd.DataFrame(data)

# ----------------------------
# Create Heatmap Data
# ----------------------------
heatmap_data = df.pivot(
    index="Genre",
    columns="Release Year",
    values="Average Rating"
)

# ----------------------------
```

```
# Plot Heatmap
# ----------------------------
plt.figure(figsize=(6, 6))

plt.imshow(heatmap_data,
           cmap="YlOrRd",
           aspect="auto")

# Axis Labels
plt.xticks(range(len(heatmap_data.columns)), heatmap_data.columns)
plt.yticks(range(len(heatmap_data.index)), heatmap_data.index)

# Color Bar
cbar = plt.colorbar()
cbar.set_label("Average Rating")

# Title and Labels
plt.title("Online Movie Rating Analysis")
plt.xlabel("Release Year")
plt.ylabel("Genre")

plt.show()
```

Output:



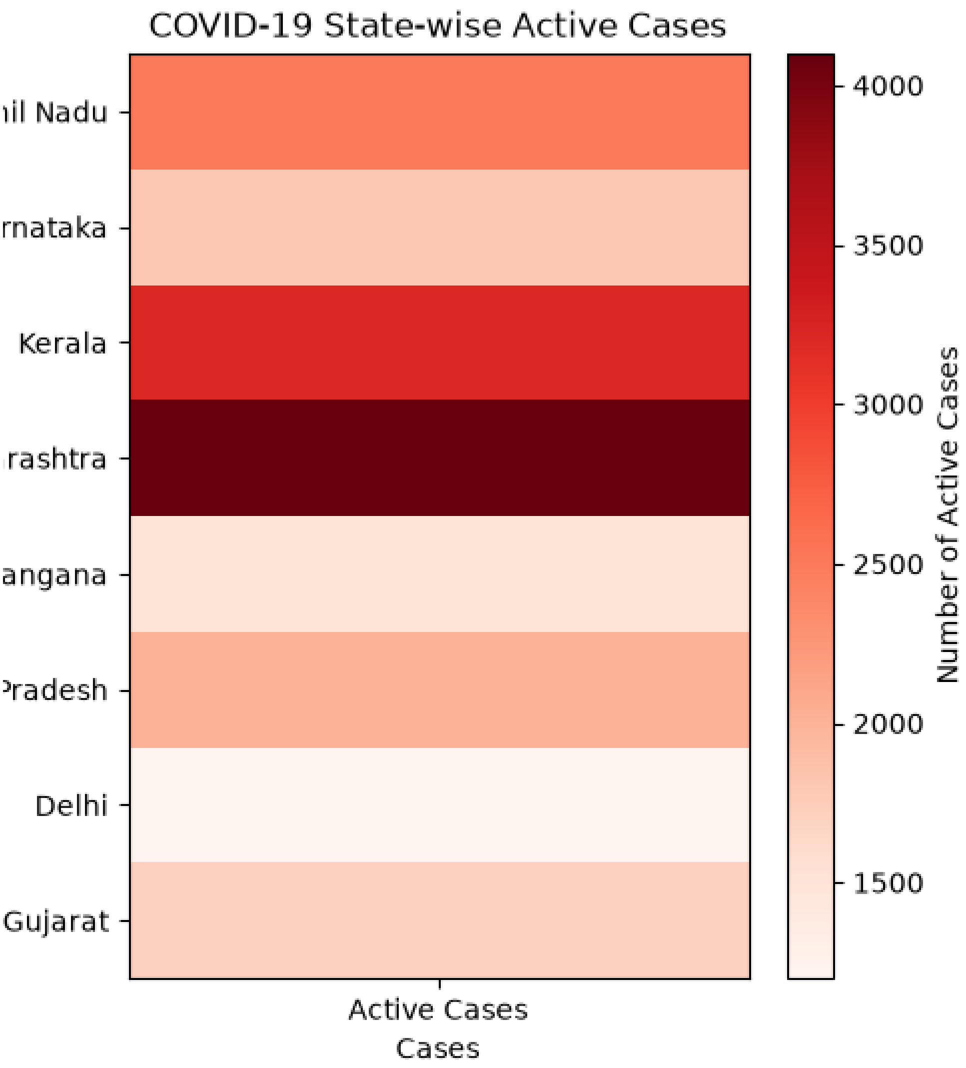
COVID-19 State-wise Analysis

Program:

```
import pandas as pd
import matplotlib.pyplot as plt

# -----------------------------
# Sample Dataset
# -----------------------------
data = {
    "State": [
        "Tamil Nadu",
        "Karnataka",
        "Kerala",
        "Maharashtra",
        "Telangana",
        "Andhra Pradesh",
```

```
        "Delhi",
        "Gujarat"
    ],
    "Active Cases": [2500, 1800, 3200, 4100, 1500, 2000, 1200, 1700],
    "Vaccination Percentage": [95, 93, 97, 94, 92, 91, 96, 90],
    "Population (Millions)": [78, 68, 36, 125, 40, 54, 20, 70]
}

df = pd.DataFrame(data)

# -----------------------------
# Create Tile Plot Data
# -----------------------------
tile_data = df.set_index("State")[["Active Cases"]]

# -----------------------------
# Plot Tile Plot (Heatmap)
# -----------------------------
plt.figure(figsize=(5, 6))

plt.imshow(tile_data,
           cmap="Reds",
           aspect="auto")

# Axis Labels
plt.xticks([0], ["Active Cases"])
plt.yticks(range(len(tile_data.index)), tile_data.index)

# Color Bar
cbar = plt.colorbar()
cbar.set_label("Number of Active Cases")

# Title and Labels
plt.title("COVID-19 State-wise Active Cases")
plt.xlabel("Cases")
plt.ylabel("State")

plt.show()
```

Output:



Iris Flower Classification

Program:

```
from sklearn.datasets import load_iris
import pandas as pd
import matplotlib.pyplot as plt

# ----------------------------
# Load Built-in Iris Dataset
# ----------------------------
iris = load_iris()

df = pd.DataFrame(
    iris.data,
    columns=[
        "Sepal Length",
        "Sepal Width",
```

```
        "Petal Length",
        "Petal Width"
    ]
)

# Add Species Names
df["Species"] = pd.Categorical.from_codes(
    iris.target,
    iris.target_names
)

# ----------------------------
# Define Colors and Shapes
# ----------------------------
colors = {
    "setosa": "red",
    "versicolor": "green",
    "virginica": "blue"
}

markers = {
    "setosa": "o",
    "versicolor": "s",
    "virginica": "^"
}

# ----------------------------
# Scatter Plot
# ----------------------------
plt.figure(figsize=(8, 6))

for species in df["Species"].unique():

    species_data = df[df["Species"] == species]

    plt.scatter(
        species_data["Sepal Length"],
        species_data["Petal Length"],
        s=species_data["Sepal Width"] * 35,   # Point Size
        color=colors[species],                # Color
        marker=markers[species],              # Shape
        edgecolors="black",
        alpha=0.7,
        label=species
    )

# ----------------------------
# Chart Customization
# ----------------------------
plt.title("Iris Flower Classification")
plt.xlabel("Sepal Length (cm)")
plt.ylabel("Petal Length (cm)")
```

```
plt.legend(title="Species")
plt.grid(True)

plt.show()
```

Output: